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Profile prediction in ECM using machine learning

Ming Wu[1], Muhammad Hazak Arshad[1], Krishna Kumar Saxena[1], Jun Qian[1], Dominiek Reynaerts[1*]

[1] *Department of Mechanical Engineering, KU Leuven & Members Flanders Make, Leuven, Belgium*

* Corresponding author. *E-mail address:* dominiek.reynaerts@kuleuven.be

Abstract

ECM (Electrochemical Machining) with a micro-second pulsed power supply can decrease stray current corrosion and thus provide better surface quality on top of higher machining accuracy. However, the use of a pulsed power supply makes it challenging to predict the final workpiece profile based on the typical multi-physical-based FEM(Finite Element method) models. First, it is impractical to introduce high-frequency pulsed electric current (ns - µs) into a multi-physical model because of time step limitations. Second, there arise problems of compatibility of the time-steps within each physical module, as e.g. hydrogen bubbles have a lifetime of ms, the power supply would use µs, and thermal phenomena take ms. Multi-scale models have been proposed, but the prediction accuracies are rather low and computation times are very long. In this article, based on machine learning approaches, we present 3 data-driven ECM models for predicting the final workpiece profile when using pulsed-ECM: the linear regression (LR) model, the neural network (NN) model, and the convolutional neural network (CNN) model. After training this data-driven ECM model with different levels of pulse voltage and electrolyte flow conditions, predictions and experimental validation are conducted. Experiments with parameters outside the training parameter window are also carried out to show the performance and general applicability of our data-driven ECM model. The machine learning model shows a good generalizing ability, the CNN model presents a prediction MSE of 7.60. The present results also demonstrate that more accurate predictions will be achieved when using in-processing data. Hence, the prediction accuracy of data-driven models can be further improved on top of advancing in-processing monitoring systems.

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Keywords: Electrochemical micromachining; pulse ECM; machine learning

Nomenclature

ECM	Electrochemical Machining
FEM	Finite Element Method
IEG	Inter electrode gap
LR	Linear regression
NN	Neural networks
CNN	Convolutional neural networks
MSE	Mean Square Error

1. Introduction

With the advances in information technology, computational methods, and advanced sensing technology; it is possible to capture and process large amounts of data. With an increasing shift to industry 4.0, the machine tools are now equipped with several advanced on-machine sensors which provide data related to the production line[1], processing status and quality[2], and machine-tool parameters[3]. Cloud computing enables proper storage of this data and makes it available as and when needed, along with edge computing, which facilitates data

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location and processing closer to the machine tool. With an enormous amount of data, it becomes complicated to process and analyze big data and deduce useful processing information.

Proper usage of these data streams can recognize highly complex and non-linear patterns in data of different types and from different sources. And developing data-driven approaches, in turn, facilitates complex and dynamic manufacturing technologies to extract useful information from big data.

Electrochemical machining (ECM) is a popular unconventional manufacturing technology to fabricate surface microstructures[4] such as micro-cavities[5], micro-protrusions[6], micro-holes[7], and micro-channels[8] on conductive and difficult-to-cut materials. It offers promising machining performance, especially in terms of high surface finish[9], no process-related tool wear, and the absence of thermal load on the workpiece. Multiphysics models[10] are widely used methods for profile prediction and hence support tool and experiments design. However, the nonlinear phenomena inherent to the ECM process such as electrolyte flow, passivation[9], bubbles, etc., make it challenging to reliably predict machined profiles. The multiphysical model[11] with well-tuned parameters can fit the experimental results beautifully for a certain processing condition but could have radically different performances in other conditions. Furthermore, it is difficult to simulate the ECM process across a wide range of processing parameters in the state-of-the-art commercial FE software because of several issues related to time-stepping, convergence, excessive mesh distortion during moving mesh, model insensitivity to a small change in parameters, etc.

The aforementioned data-driven models[3], on the other hand, present the capability of adapting to various processing parameters and associated nonlinear phenomena for a manufacturing process, thereby providing a more generalized model for profile prediction.

In this article, we present a framework of a data-driven model for the profile prediction of machined features with the ECM process. Different supervised machine learning models are used to investigate the relationship between the dimensions of cavities and the applied pulsed current, voltage, electrolyte flow rate, and environmental temperature. The applied pulsed current and temperature are monitored and recorded during the machining process. These machine learning models are evaluated by an out-of-domain evaluation dataset, which applies processing parameters that are never been fed to the data-driven models and also investigates the feasibility of using data-driven models over a wide range of the parameter set. The advantages of using a data-driven model with simple training with processing parameters are investigated.

2. Materials and Methods

2.1. Experimental Setup and methodology

The experiments for collecting the data were conducted on an in-house developed hybrid laser-electrochemical machining (LECM)[12]. The setup, as shown in Figure 1, can execute complex 2D machining trajectories through the synchronized activation of XY-stage motion (LUMS software) and microsecond pulsed voltage source (LabVIEW® MMI on NI® cRIO 9045). A stainless-steel tubular electrode (1.2 mm outer diameter and 0.35mm internal diameter) was used for machining experiments. The electrolyte was supplied through a 3D printed electrolyte guide with internal flow channels. The material is PLA (Polylactic acid) and the flow channel is embedded inside the guide to avoid electrolyte flow towards the spindle and force it to flow towards the tool electrode.

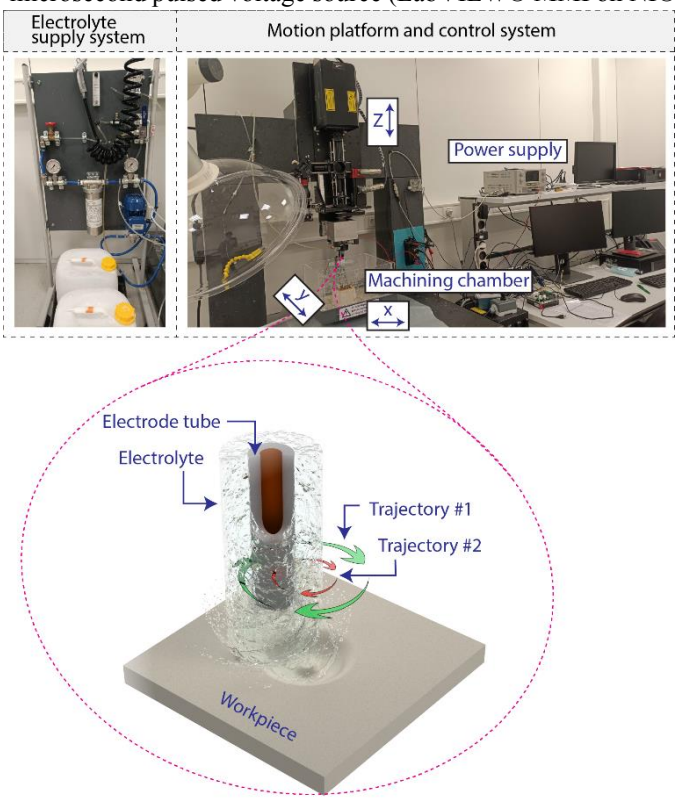

Figure 1: Experimental setup

The electrolyte used was 20% by weight aqueous sodium nitrate.

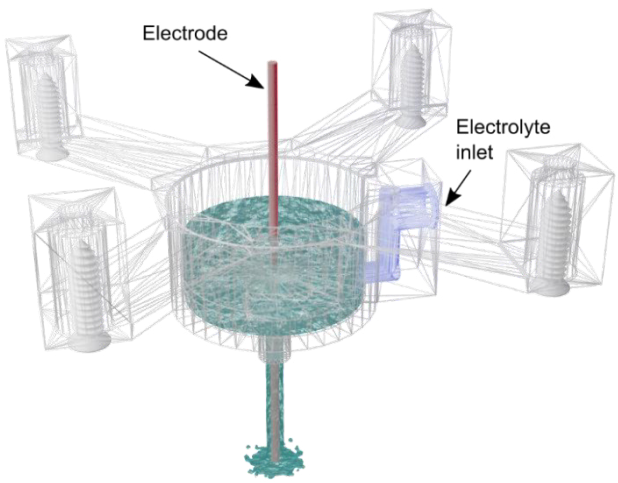

Figure 2: 3D printed electrolyte guide

2.2. Data Collection

Cavities were machined in two concentric circles trajectory as shown in Figure 1 with outer radius 62.5 µm (trajectory #1) and inner radius 12.5 µm (trajectory #2) in two repetitions. The two concentric circles trajectory were used to improve electrolyte flow into the extremely small IEG of 80um. The dimensions of the trajectory were such small that they were less than 1% of the internal diameter (0.65 mm) of the tubular electrode tool with a diameter of 1.2 mm. Thus electrochemical reactions following these trajectories could represent point removal but with dynamic flushing conditions.

Average peak machining current and temperature (4 temperature sensors were implemented: 1 in the electrolyte tank, 2 distributed in the lab to check the room temperature, and 1 attached in the downstream side of the machining area) were logged every second through the NI-9223 module. The machined samples were cleaned in an ultrasonic bath (DI water of 8 MΩ.cm) for 20 mins before acquiring the cavity profiles using the Sensofar® S Neox 3D Optical Profiler.

Inconel 718 was used as a workpiece for machining the cavities due to its wide range of applications in the aerospace industry. All experiments were done at a fixed inter-electrode gap (IEG) of 80µm and machining time of 14s.

As shown in Table 1, the processing parameters, i.e., voltage, electrolyte fflow rate, pulse duration, and duty cycle, are varied. The applied pulse current and temperature are monitored and recorded during the machining process. The training dataset is built for training the machine learning algorithms to predict the machining profile. A second dataset, i.e., the validation dataset, is recorded for evaluating how well these models are truly generalizing to the prediction and not simply overfitting the training dataset. Experiments are carried out with processing parameter combinations of all the values in Table 1. For each combination, 5 cavities are machined. Therefore, the training dataset should contain: $3^4 \times 5 = 405$ cavities. However, the cavities can hardly be machined properly with the lowest and highest flow rate because of the insufficient electric conductivity and electrolyte stability in our case. A large portion of cavities machined with the flow rate of 0.14 or 0.95 ml/s was dropped from the legit dataset. Finally, 140 cavities were collected for the training dataset, and $2^2 \times 5 = 20$ cavities were collected for the validation dataset.

Table 1 Processing parameters for training and validation

	Training			Validation	
Voltage (V)	15	20	25	22.5	30
Flow rate (ml/s)	0.14	0.47	0.95	0.47	
Pulse duration (µs)	8	20	50	20	50
Duty cycle (%)	20	30	50	50	

As shown in Figure 3, two types of cavities are produced with different processing parameters. The profile features, including the a) diameter *D* and b) depth of the cavity *H*, the c) diameter *H_D* and d) height of the center hill *H_H*, are measured as outputs for machine learning models. The cross-sectional profiles of cavities in the following sections are interpolated by these features as well.

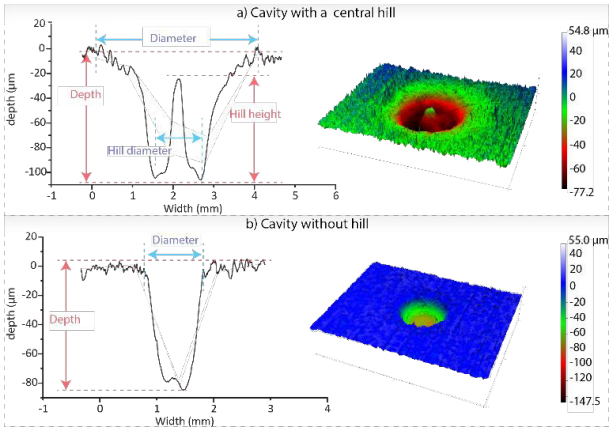

Figure 3: Different shapes of cavities observed in the ECM process with a tubular tool. a) cavity with a central hill; b) cavity without a central hill.

2.3. Machine learning models

For predicting the dimension of cavities, three machine learning models are employed.

1. *The linear regression model.*

Linear regression is a simple machine learning model with good interpretability. A linear regression model is established, as shown in Figure 4, to predict the cavity features listed in Table 2. The prediction outputs $\boldsymbol{y}$ to the inputs $\boldsymbol{x}$ can be given as:

$$y(x) = w_0 x_0 + w_1 x_1 + \cdots + w_i x_i + b \quad (1)$$

where, w_i is the weight corresponding to the feature x_i and b is the bias.

In the linear regression model, the processing parameters and the recorded signal (the mean value during the processing time) can be feature-fused, together as the input data. The linear regression model can also use the process parameters or the recorded signal independently as the input data.

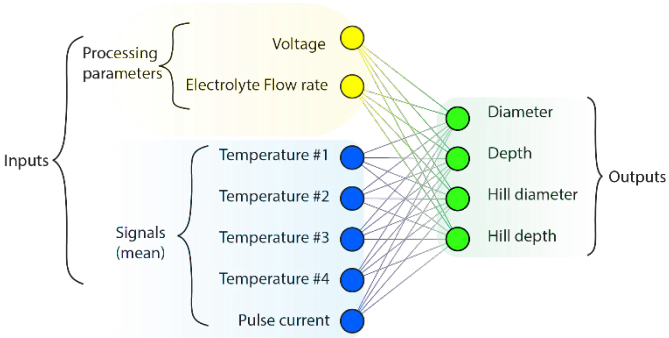

Figure 4: The linear regression model applied to the ECM process.

2. *The neural networks model.*

The neural networks model (NN) is a classic neuron-based machine learning model that consists of multiple hidden layers.

This neuron is a computational unit with input $\boldsymbol{x}$ operated by matrices of weights $\boldsymbol{W}$ and bias $\boldsymbol{b}$, the output for each neuron $h_{w,b}$ is calculated by:

$$h_{W,b} = f(W^T x) = f(\sum W_i x_i + b) \quad (2)$$

For electrochemical processes, there are many nonlinear equations. The hidden layer in the neural network model can describe nonlinear phenomena. In this paper, as shown in Figure 5, a neural network with 3 hidden layers is set up.

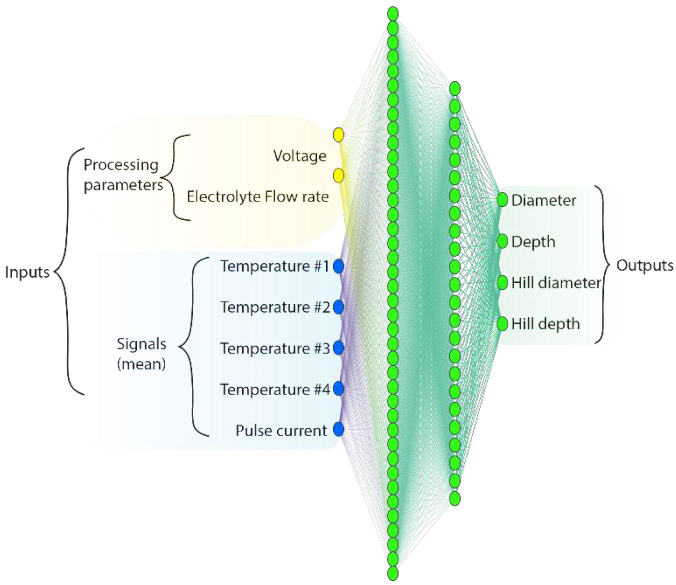

Figure 5: The neural network model.

Similar to the linear regression model, the neural network model can use the process parameter or the recorded signal (the mean value during the processing time) separately, along with fusing them, as input data.

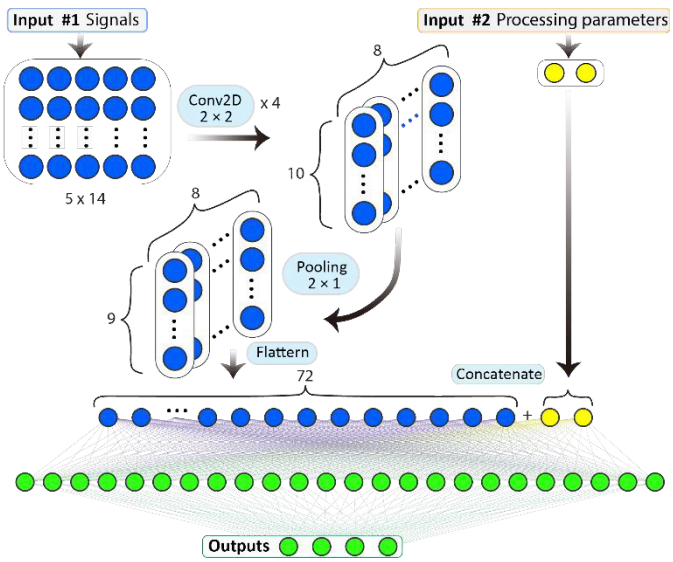

Figure 6: The convolutional neural network model.

3. The convolutional neural networks model.

Convolutional neural network (CNN) is one of the typical algorithms of deep learning, including convolution layers and pooling layers, as shown in Figure 6. Instead of using the mean value of the recorded signal during the processing time, CNN can use all the data through the recording time as inputs. The size of the input signal hence will be 5×14, where "5" presents the log channels, 1 from the electric current and 4 from the temperature, and "14" presents the time series size.

The feature map $\boldsymbol{F}$ can be defined by the convolution kernel $\boldsymbol{K}$ and input matrix $\boldsymbol{I}$ as $F = I \circledast K$, where $\circledast$ denotes the convolutional operation, the feature map can be regarded as a collection of features produced by the kernel $\boldsymbol{K}$, which means the convolution kernel $\boldsymbol{K}$ extract certain information from the input matrix $\boldsymbol{I}$.

Therefore, a convolutional model with a convolution kernel size of 2×2 is developed so that its feature maps should cover the information both horizontally, which means features collection cross the record channel, and vertically, which means feature collection cross the sampling time.

All the machine learning models are deployed with Tensorflow backend and solved on a Nvidia® GPU RTX 2070. With the presented data-driven approaches for profile prediction in ECM, the computational time only costs less than 1s to make predictions for all the validation datasets.

The MSE is computed from the squared error between the predicted ($\hat{y}_i$) and observed (y_i) measurements for each j-th experimental run of the validation dataset, as shown in equation (3) Here $\boldsymbol{n}$ is the total number of experimental runs in the validation dataset.

$$MSE(y, \hat{y}) = \frac{1}{n}\sum_{i=0}^{n-1}(y_i - \hat{y}_i)^2 \tag{3}$$

In this paper, the information is monitored and collected in different ways. It is hard to decide initially whether this information is redundant or essential for profile prediction. The following sections will analyze the predictive capabilities of different machine learning models.

3. Results and discussion

3.1. Prediction accuracy of machine learning models

The predicted MSEs of different algorithms with different inputs are shown in Figure 7 and Table 2, where P denotes the processing parameters and circles in Figure 7 denote the outliers; S means the recorded signal data during the process. For LR and NN models, S means the time average value of the recorded signal. For the CNN model, S means the raw recorded data.

Table 2 Overall MSE of different machine learning methods in the validation dataset

	LR			NN			CNN	
	P	S	P+S	P	S	P+S	S	P+S
D	1.20	0.30	0.71	0.72	0.92	0.92	0.74	0.42
H	17.8	18.0	9.1	12.6	9.2	14.8	9.4	13.0
H_D	0.48	0.26	2.40	0.37	0.43	0.54	0.12	0.09
H_H	19.6	19.4	23.1	22.8	21.2	20.8	20.3	16.9
Total	9.76	9.51	8.82	9.12	7.93	9.28	7.64	7.60

There are several nonlinear phenomena, including electrochemical reactions, in the ECM process. Because of the inability to learn and express the nonlinear relationships, it can be seen that the linear regression (LR) model has higher

prediction errors and depicts poor prediction accuracy than the CNN model. After the feature fusion is performed as inputs for the linear regression model sometimes, the error is even higher. This indicates that using recorded signals may even be held as noise for linear regression.

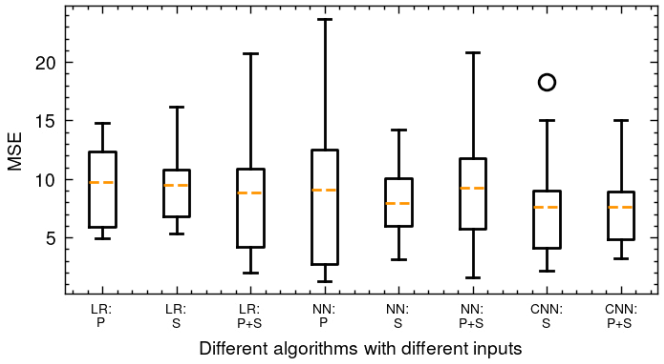

Figure 7: Comparison of different algorithms with different inputs

For the neural network (NN) model with different inputs, when using the recorded signal solely as the input data, the average prediction MSE is 7.93; whereas, when using a combination of processing parameters and the mean value of recorded signals, the average prediction MSE is 9.28, the prediction error increased. This presents the NN model's limits of absorbing different information.

Since the processing parameters cannot be individually considered as inputs for the convolutional neural network (CNN) model, the results of fused-feature input data sets are presented.

It can be seen that, when using fused feature inputs, the CNN model represents the lowest prediction MSE, i.e. the best performance. This indicates that CNN has a stronger ability to analyze different data and adapt to nonlinearity. While the LR and NN models can only use recorded signal data by calculating their mean value, the CNN model can work with record data with no subtraction. The more information fed to the machine learning model, the better prediction accuracy it can present.

3.2. Data-driven approaches in the unsatisfactory processing conditions

In the classical ECM model, all the analysis is based on ideal and fixed machining conditions during the processing time, which is heavy to be achieved and maintained in practice.

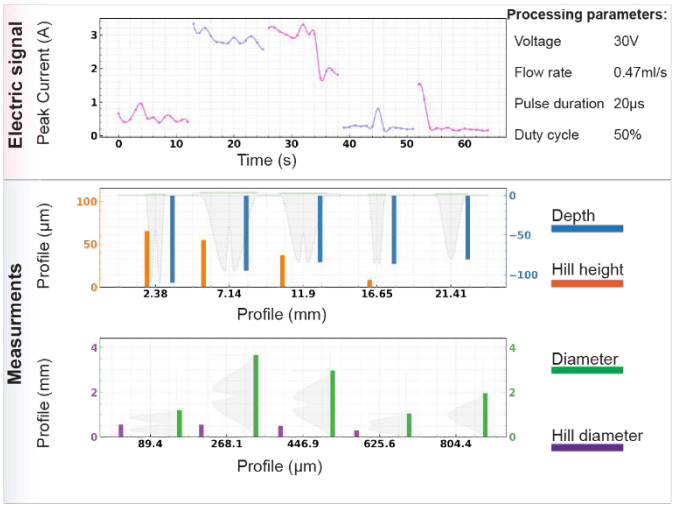

Figure 8: Machined cross-sectional profile with: voltage 30V, flow rate 0.47ml/s, pulse duration 20μs, duty cycle 50%

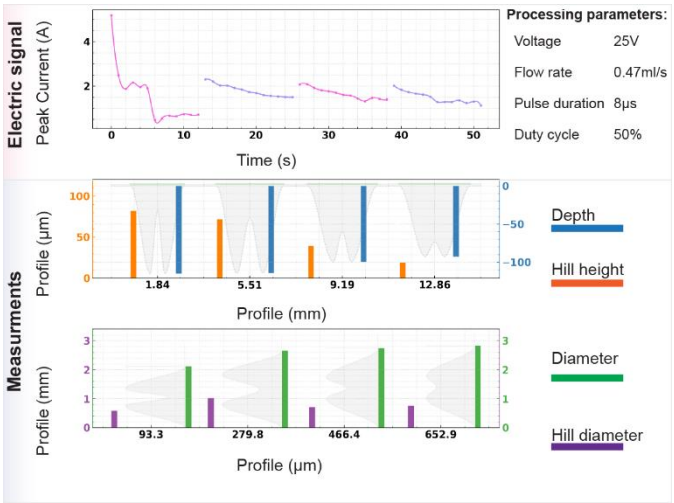

Figure 9: Machined cross-sectional profile with: voltage 25, flow rate 0.47ml/s, pulse duration 8μs, duty cycle 50%

As shown in Figures 8, 9, the cavities in each figure were machined in a certain processing condition but presented profoundly different profiles. In the meantime, a significant difference can be clearly noticed from the electric signal record. This suggested some missing information that is essential to the ECM processing could be found in the recorded data.

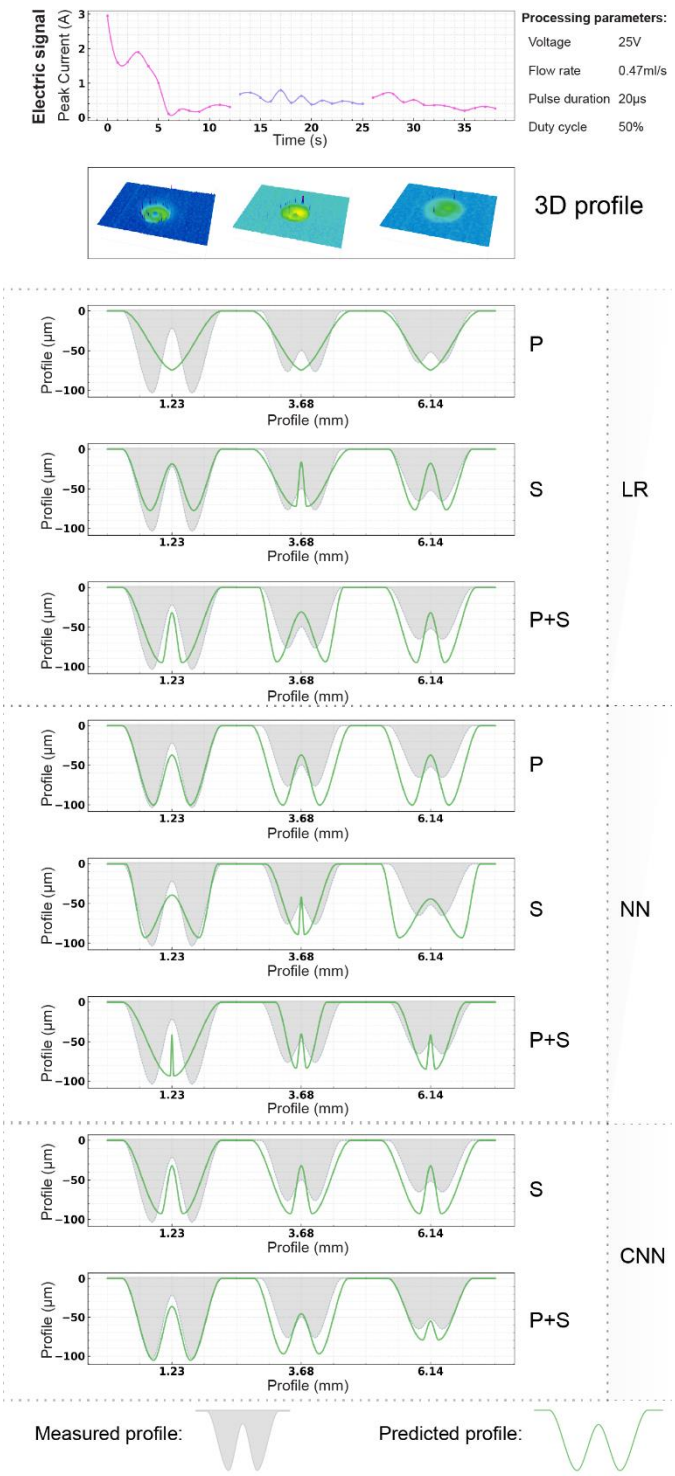

Figure 10: Predicted profile of different algorithms with different inputs

As shown in Figures 10, 11, the identical processing conditions can cause different profiles, and different algorithms with different inputs showed a different predicted accuracy. When only using the processing parameters as inputs, identical predictions will be presented, where is far from the ground truth. When using the recorded signal data as inputs or one part of inputs, the predictions become different, as the signal data is different.

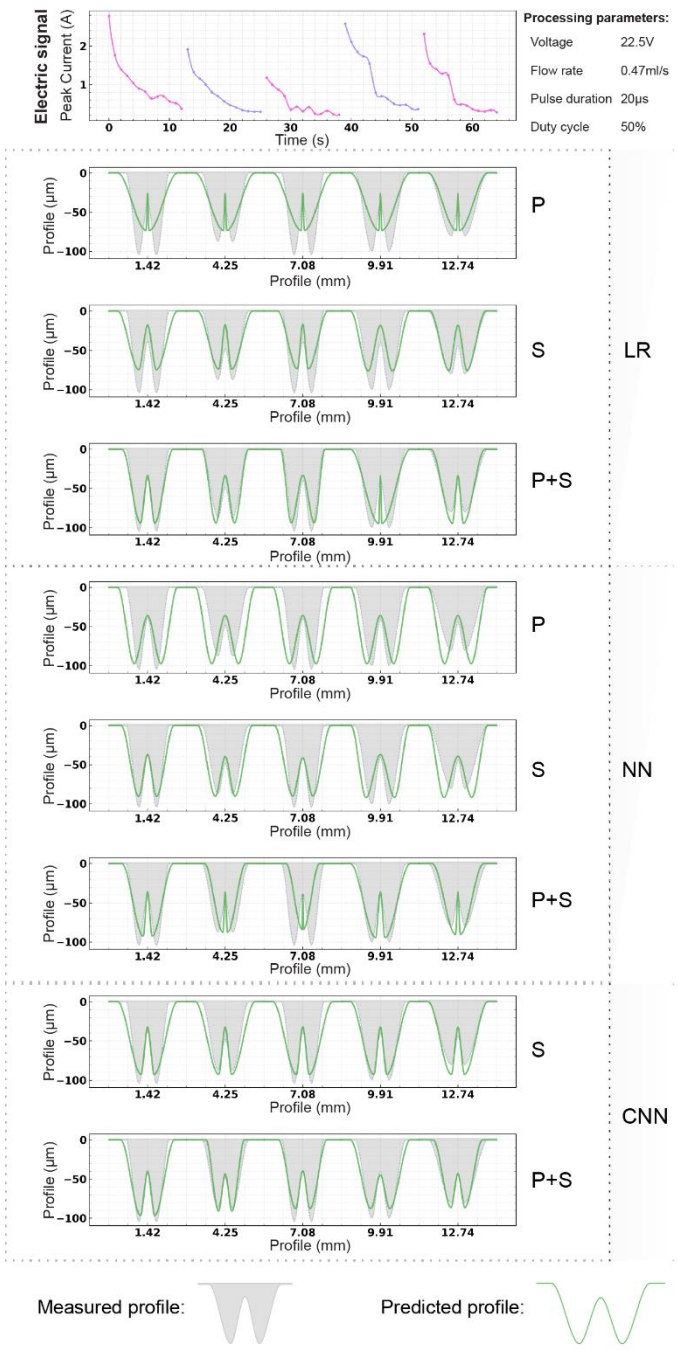

Figure 11: Predicted profile of different algorithms with different inputs

It can be seen that the predictions given by the CNN model with P+S inputs fit the measurements most. In addition, this confirms the results of Section 3.1 that the data-driven approaches with more complex architectures perform better; the inputs with more information fed to the data-driven model perform better.

3.3. Limitations of machine learning models

Figure 12 demonstrated the predicted MSE for different profile features. It can be seen that the diameter of the machined cavity and the center hill have lower predicted errors, whereas the prediction error of the cavity's depth and the center hill's height is unexpecting high.

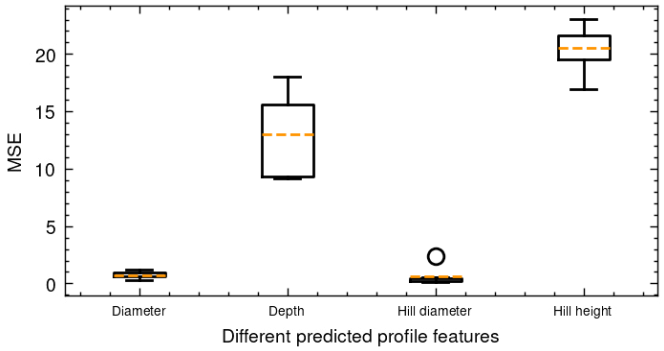

Figure 12: Predicted MSE for different profile features

One plausible reason is that the information collected now is not enough for data-driven approaches to faithfully predict the cavity's depth or the center hill's height. For instance, as the in-processing electrode flow condition in the IEG is currently beyond reach, whereas there is a consensus that the flow field in the inter-electrode gap (IEG) is crucial to the machining process[11] and is sensitive to the bubbles generation and machined workpiece surface profile, the effects of electrochemical reactions principally influenced by the electrolyte are not perceived by the machine learning models, hence the faithful predictions which cover all the processing scenarios in this investigation are practically challenging to be presented. This is the inherent imperfection of data-driven intelligence — no data, no intelligence.

4. Conclusion

1. The machine learning model shows a good generalizing ability. When using the out-of-domain datasets, the CNN model presents a prediction MSE of 7.60.

2. With data recorded during the machining process, more accurate results will be achieved, however not all the algorithms can get the same advantages when adopting more information. The extra information could be noise or irrelevant inputs for the LR and NN model.

3. Despite the unsatisfactory processing conditions, good predictions can still be presented because of the employment of recorded in-processing data.

4 The predicted accuracy is relatively low for the cavity's depth or the center hill's height, which may be caused by the IEG electrolyte flow condition and requires further investigation on the in-processing monitoring system.

Acknowledgments

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